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SHORT-PAPER

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ExaRanker: Synthetic Explanations Improve Neural Rankers

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ABSTRACT

Recent work has shown that incorporating explanations into the output generated by large language models (LLMs) can significantly enhance performance on a broad spectrum of reasoning tasks. Our study extends these findings by demonstrating the benefits of explanations for neural rankers. By utilizing LLMs such as GPT-3.5 to enrich retrieval datasets with explanations, we trained a sequence-to-sequence ranking model, dubbed `ExaRanker`, to generate relevance labels and explanations for query-document pairs. The `ExaRanker` model, finetuned on a limited number of examples and synthetic explanations, exhibits performance comparable to models finetuned on three times more examples, but without explanations. Moreover, incorporating explanations imposes no additional computational overhead into the reranking step and allows for on-demand explanation generation. The codebase and datasets used in this study will be available at https://github.com/unicamp-dl/ExaRanker

CCS CONCEPTS

• **Information systems → Novelty in information retrieval**; • **Computing methodologies → Neural networks**.

KEYWORDS

Few-shot Models, Large Language Models, Generative Models, Explanations, Synthetic Datasets, Multi-stage Ranking

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1 INTRODUCTION

Pretrained transformers such as BERT [9] and T5 [35] finetuned on hundreds of thousands of examples brought significant gains to information retrieval (IR) tasks [12, 14, 16, 17, 22, 24, 26, 27, 31, 44, 51, 59, 61, 65]. When queries and documents from a task of interest are similar to the ones in the finetuning data, it is likely that a model will have an effectiveness better than that of unsupervised models. For example, a monoT5 [32] reranker finetuned on 400k positive query-passage pairs from MS MARCO [1] outperforms BM25 [39] in 15 of 18 datasets of the BEIR benchmark [40, 41].

However, when the number of labeled examples is limited, the effectiveness of the model decreases significantly. For example, a BERT reranker finetuned on a mere 10k query-relevant passage pairs performs only slightly better than BM25 on the MS MARCO passage ranking benchmark [32]. Increasing the model size [41] or pretraining it on IR-specific objectives [13, 19] alleviate the need for more finetuning data, at the expense of increased computational resources.

We argue that one reason neural retrievers need extensive amounts of training examples is that they are finetuned on categorical labels (e.g., true/false). These labels provide limited information about the task to be learned, making it more challenging for the model to grasp the nuances of the task. For example, imagine attempting to teach a human to evaluate the relevance of passages to queries, but only being able to communicate the words "true" or "false" per query-passage pair. The learning process would be more efficient if explanations in natural language were provided to explain why a passage is relevant or not to a given query.

In this work, we propose a method for training retrieval models using natural language explanations as additional labels, which reduces the need for training examples. Furthermore, we show that few-shot LLMs such as GPT-3.5 [34] can be effectively employed to automatically augment training examples with explanations, enabling IR practitioners to apply our method to other datasets without the need for manual annotation. Our findings indicate that the utility of incorporating explanations decreases as the number of training examples increases. Furthermore, our experimental results demonstrate that finetuning a model to generate a label prior to an explanation leads to superior performance in comparison to generating an explanation before the target label. This outcome may be counterintuitive and contradicts previous findings in chain-of-thought works.

2 RELATED WORK

Recent research has shown that inducing a language model to generate step-by-step rationales in natural language improves performance in various tasks that require reasoning [5, 10, 18, 21, 28, 33, 37, 60, 63, 64]. However, experiments using induced explanations are typically carried out on models with billions of parameters, which are impractical to use in information retrieval tasks. For example, reranking 100 passages for one query using a model with

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175B parameters would take at least one minute on four A100 GPUs. In this work, we propose a method to distill the knowledge from these large models and effectively use it to improve the quality of results in a ranking task.

Our method is related to InPars [3, 20], Promptagator [8], and UPR [43], with the distinction that we use LLMs to augment existing target labels from training datasets with additional information relevant to the task at hand, rather than generating or scoring queries from passages. We view these methods as complementary to our method, and anticipate potential integration of these approaches in future research.

A large body of work addresses the task of generating explanations for a ranked list of results [11, 36, 42, 45–49, 53, 56, 57, 62, 66]. For example, GenEx [36] generates noun phrases (e.g., "impacts on medicare tax") for a given query-document pair. Snippet generation can also be regarded as providing explanations for why certain results are being presented to the user [2, 6, 54, 55]. A key distinction of these works in comparison to ours is that our goal is to use explanations as a means to improve the effectiveness of a retrieval model, rather than solely for the purpose of explaining a list of results to users. This is similar to the work of Sheshera et al. [30], who use naturally-occuring co-citations in scientific corpora as a enhancing signal to train document similarity models. We do not claim that our method makes a retriever interpretable as we have not evaluated the correctness of the generated explanations. Our goal here is orthogonal to that of building interpretable retrievers: we show that explanations improve the effectiveness of retrievers. Nevertheless, our method may help users to understand a ranked list of results, but further examination of this aspect is left for future research.

3 METHODOLOGY

The proposed method is descrived next. It starts by generating explanations for query-passage-label triples using an LLM model with in-context examples. These training triples are then augmented with the generated explanations, and a sequence-to-sequence model is finetuned to produce the target label followed by the explanation. In the inference phase, the finetuned model is used to estimate the relevance of a query-passage pair, based solely on the probability assigned to the label token.

We begin by randomly selecting 15k pairs of (query, relevant passage) and another 15k pairs of (query, non-relevant passage) from the training set of the MS MARCO passage ranking dataset, and then generate explanations for these 30k pairs. Manually generating explanations for such a large volume of text would be cost-prohibitive, so we use `text-davinci-002`[1] with a few-shot prompt to infer explanations for query-passage-label triples. Earlier versions such as `text-davinci-001` or smaller models such as `curie` did not produce reasonable explanations. We use greedy decoding, and the output is limited to 256 tokens. The few-shot prompt, illustrated in Figure 1, contains 7 examples that were selected from the MS MARCO training dataset, and has an average of 1400 tokens, including the 256 tokens of the explanation to be generated. As of February 2023, generating an explanation for each query-passage-label triple costs 0.028 USD using the OpenAI API. Due to cost constraints, the dataset was limited to 30k examples, which amounts to 840 USD.

[1]beta.openai.com

Prompt - text

Instruction: explain if the passage is relevant to the question.
###
Example 1:
Question: What is a thyroid nodule

Passage: 1 Thyroid cancer: An uncommon form of cancer, thyroid cancer is usually curable. 2 Surgery, radiation, and hormone treatments may be used to treat thyroid cancer. 3 Thyroid nodule: A small abnormal mass or lump in the thyroid gland.

Final Answer: The passage is relevant to the question

Explanation: The question is what is a thyroid nodule. The last sentence of the passage describes the thyroid nodule.

...
###
Example 8:
Question: **{query}**

Passage: **{passage}**

Final Answer: The passage is **{"not " if label==False else ""}**relevant to the question

Explanation: **{explanation}**

Figure 1: Prompt used to generate explanations for a query-passage-label triple (presented in Python's f-string notation).

Note that we also use the label (relevant or not relevant) in the prompt and request only the explanation for the triple query-passage-label. In preliminary experiments, we noticed that the model generates better explanations when the label is provided, as it is not confounded by the classification task and can focus specifically on the explanation.

Once the 30k explanations are generated, a sequence-to-sequence model is finetuned using the following input/output templates (shown using Python's f-string notation):

Input: `Is the question {query} answered by the {passage}? Give an explanation.`

Output: `{label}. Explanation: {explanation}`

The terms `{query}` and `{passage}` are the query-passage pair extracted from the MS MARCO dataset. The `{label}` is `true` if the passage is relevant to the query and `false` otherwise. Finally, `{explanation}` is the one generated by the LLM as explained before.

The T5-base model was used as the starting point for the finetuning phase. The model was finetuned for 30 epochs using the AdamW optimizer [25] with a learning rate of 3e-5, weight decay of 0.01, and batch size of 128 examples (64 positives and 64 negatives). The maximum number of input and output tokens were each limited to 512. Sequences exceeding these values were truncated during both training and inference. For comparison purposes, a T5-base model was also finetuned using the same hyperparameters and dataset, but without the inclusion of explanations in the target text.

After finetuning, we evaluate the models on a passage ranking task. The input to the model is the same as presented before, and, as a consequence, it generates the same output pattern.[2] The probability of the first output token is used as the relevance score s for the query-passage pair:

$$s = \begin{cases} 1 + p_0, & \text{if } t_0 = \text{true} \\ 1 - p_0, & \text{if } t_0 = \text{false} \\ 0, & \text{otherwise} \end{cases}$$

[2]In our experiments the model always generates an output that matches the target template.

where t_0 is the token generated in the first decoding step and p_0 is the probability assigned by the model to that token, i.e., the probability from the softmax after the logits.

We limit the model to generate only the label for the query-passage pair (i.e., the first token) and omit the full explanation text to save processing time. Since the model was trained with causal mask, that is, only tokens generated so far influence the prediction of the next token, the relevance scores are the same had the model generated an explanation. Alternatively, explanations can be generated by decoding more tokens until the termination token (e.g., <EOS>) is generated.

To evaluate the benefits of explanations in a more realistic scenario in which training data is not available, we evaluate the model in a zero-shot manner on 6 datasets from the BEIR benchmark [52]: Robust04 [58], TREC-COVID [38], DBPedia [15], FiQA [29], TREC-NEWS [50] and NFCorpus [4]. TREC-DL 2020 [7] is used as a validation set to select the best checkpoint and hyperparameters.

4 RESULTS

The results are presented in Table 1. The first two rows are the BM25 "flat" and "multifield" baselines from Pyserini [23]. In "flat" indexes, document titles and contents are concatenated and stored as a single field while "multifield" indexes store them as separate fields. The third row is a monoT5-base model finetuned on 400k positive query-passage pairs from the MS MARCO dataset for one epoch.[3] All rerankers reported in this work rerank 1000 documents retrieved from BM25's "flat" indexes. In our analysis, we concentrate on the number of positive query-passage pairs, which require manual annotation and thus represent a costly task when developing a search engine. On the other hand, negative query-passage pairs can be readily obtained through automatic selection using a retriever once the queries have been collected. While we still need to generate explanations for negative query-passage pairs, we anticipate that this cost will decline in the future as open-source LLMs become more widely available.

Compared to the BM25 baselines, the finetuned models have a much higher nDCG@10 scores. In the fourth and fifth rows, we compare models finetuned using the same hyperparameters and examples, but without explanations (monoT5) and with explanations (ExaRanker). The ExaRanker model outperforms the model without explanations in all 7 datasets. On average, it improves the score by 0.8 nDCG@10 points in the zero-shot evaluation. This result shows the benefits of using the explanation to provide more data and reasoning during the training phase, and enable the model to succeed with much less training data.

To better understand the benefits of using explanations, we have reproduced the training methodology with smaller datasets: 20k, 10k and 5k examples. All datasets have an equal number of positive and negative query-passage pairs.

The ExaRanker model outperforms the model finetuned without explanations in all cases. When finetuning on 15k positive examples, ExaRanker performs 0.8 points better than monoT5. When finetuned on 10k examples, the improvement is 1.1 points and increases to 2.6 when finetuned on 5k positive examples. However,

[3]https://huggingface.co/castorini/monot5-base-msmarco-10k

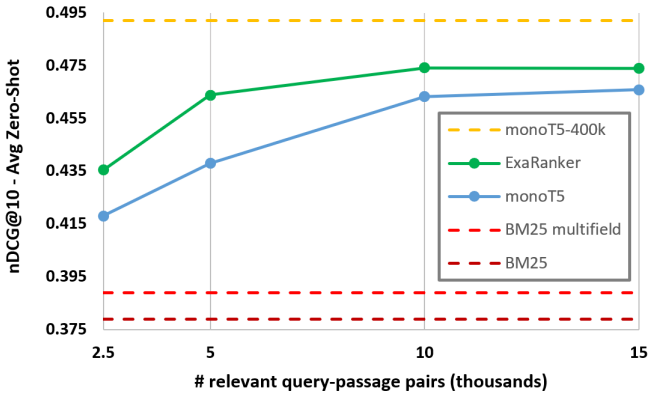

Figure 2: Average zero-shot results on 6 datasets of the BEIR benchmark. monoT5-400k is finetuned on the 400k relevant query-passage pairs from MS MARCO without explanations.

ExaRanker finetuned on 15k and 10k positive examples are on average 1.7 nDCG@10 points behind monoT5-400k, which has been finetuned on 400k positive examples. We were unable to conduct a more equitable evaluation of this model since generating explanations for 400k positive and 400k negative examples would incur a cost of 10,000 USD, exceeding our allocated budget. Nevertheless, this amount may be reasonable for certain research laboratories that are willing to carry out such an experiment.

Figure 2 provides a visualization of this trend. We see that the improvement in using explanations decreases as the number of samples increases. Comparing the monoT5 results finetuned using 15k positive pairs with ExaRanker using 5k positive pairs, we see that the average scores are very close but ExaRanker uses 3x less data for finetuning. These results suggest that data augmentation through explanations is an effective way to distill knowledge from LLMs.

4.1 Qualitative Analysis

Table 2 presents some sample outputs for a qualitative comparison of the predictions made by ExaRanker on the TREC-DL 2020 dataset. The model is able to generate reasonable explanations: in the second example, it correctly predicted the relevance of the passage, even though it only mentions specific terms from the query, such as "early pregnancy". However, it may struggle with more fine-grained relationships, such as identifying the relationship between electromagnetic and radio waves, as shown in the third example. Overall, explanations appear to support a better understanding of the relationship between the query and the passage, both quantitatively and qualitatively.

4.2 Ablation Experiments

Two additional experiments were conducted to better understand the contribution of adding explanations as training targets.

First we finetuned another ExaRanker model for 30 epochs, with the same hyperparameters but inverting the order in which label and explanations are generated: first an explanation is generated, then a relevance label. It is reasonable to expect that this order

Table 1: Main results (nDCG@10), including average zero-shot (all except DL 20). The column "Ft Pos." is the number of positive training examples on which the model was finetuned. In BM25 multifield, document titles are separated from content.

Method	Ft Pos.	DL 20	Robust	Covid	Dbp	FiQA	News	NFC	Avg ZS
BM25 flat	-	0.478	0.407	0.594	0.318	0.236	0.395	0.321	0.379
BM25 multifield	-	-	0.407	0.656	0.313	0.236	0.398	0.325	0.389
monoT5	400k	0.652	0.536	0.777	0.419	0.413	0.447	0.357	0.491
monoT5	15k	0.656	0.523	0.746	0.392	0.382	0.409	0.344	0.466
ExaRanker	15k	0.680	0.531	0.747	0.394	0.403	0.417	0.351	0.474
monoT5	10k	0.643	0.510	0.749	0.379	0.374	0.426	0.341	0.463
ExaRanker	10k	0.667	0.527	0.752	0.409	0.393	0.418	0.347	0.474
monoT5	5k	0.625	0.488	0.693	0.364	0.337	0.417	0.328	0.438
ExaRanker	5k	0.665	0.505	0.750	0.389	0.380	0.414	0.345	0.464
monoT5	2.5k	0.611	0.486	0.666	0.334	0.328	0.370	0.325	0.418
ExaRanker	2.5k	0.650	0.496	0.686	0.393	0.306	0.398	0.335	0.436

Table 2: Outputs generated by the ExaRanker model.

Input	Output	Target
Is the question: "average salary for dental hygienist in nebraska" answered by the document: "Dental Hygienist Salaries by Area. According to the Bureau of Labor Statistics, Dental Hygienists in the state of Nebraska can expect to earn an average of $65,236 annually or $31.36 per hour."? Give an explanation.	true. Explanation: The question is about average salary for dental hygienist in nebraska. The passage describes the average salary for dental hygienist in Nebraska.	true
Is the question: "can fever cause miscarriage early pregnancy?" answered by the document: "Pregnancy Early symptoms of pregnancy include a missed period, morning sickness, breast swelling, fatigue, and more. Chickenpox (varicella) Chickenpox is an illness that causes a red, itchy rash, fever, headache, sore throat, and body aches"? Give an explanation.	false. Explanation: The question is about if fever can cause miscarriage early pregnancy. The passage is about early symptoms of pregnancy and chickenpox.	false
Is the question: "what medium do radio waves travel through" answered by the document: "An electromagnetic wave (this includes light) does not require a physical medium - it can travel through empty space."? Give an explanation.	false. Explanation: The question is about what medium do radio waves travel through. The passage is about electromagnetic waves and how they travel through empty space.	true
Is the question: "do google docs auto save" answered by the document: "Inter-sheet formulas in Google Docs are similar to those used in Microsoft Excel. Using this ability, you can post a value automatically in a cell on one tab that is calculate or posted in a cell on another tab."? Give an explanation.	true. Explanation: The question is about if google docs auto save. The passage describes how google docs auto save.	false

Table 3: Ablation of the output template. Results on TREC-DL 2020 with models finetuned on 15k pos. + 15k neg. samples.

Output template	nDCG@10
`{label}. Explanation: {explanation}`	0.680
`Explanation: {explanation}. {label}.`	0.584

Table 4: ExaRanker results from a monoT5-base model finetuned on 400k positive examples from MS MARCO.

Model	Avg ZS
monoT5 (ft on 400k pos)	0.514
ExaRanker (from monoT5)	0.512

would lead to better results, as it induces the model to elaborate on the relationship between query and passage prior to predicting a label.

However, as shown in Table 3, the nDCG@10 score dropped 9.6 points compared to generating a relevance label followed by an explanation. We believe that this behavior results from the challenge of consolidating the probabilities of multiple generated tokens into a single ranking score. We tried different approaches, such as using the average token probabilities of the entire sequence or using only the probability of the final token (`true` or `false`). In all cases, nDCG@10 was lower than the method described in Section 3.

In a second ablation study, we tested if explanations would improve a model already finetuned on a large ranking dataset. For that, we further finetuned on 15k positive + 15k negative examples with explanations a monoT5-base model already finetuned on 400k positive pairs from MS MARCO. We used the same hyperparameters as described above. Results in Table 4 show a negligible difference of 0.2 nDCG@10 points. As explained before, the benefits of explanations seems to be reduced when a large training dataset is available. The experiment also shows that finetuning on explanations does not compromise the effectiveness of a ranker, while also equipping it with the ability to generate explanations.

5 CONCLUSION

We have proposed a method for training neural rankers using natural language explanations as additional labels. Our results demonstrate that the use of explanations can significantly reduce the need for training examples, particularly when fewer labeled examples are available.

We also showed that large language models can be used to effectively generate these explanations, thus allowing our method to be applied to other IR domains and tasks. Importantly, our method does not significantly increase the time required to rerank passages as only the true/false token is used during inference.

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